Preserve and Personalize:
Personalized Text-to-Image Diffusion Models without Distributional Drift

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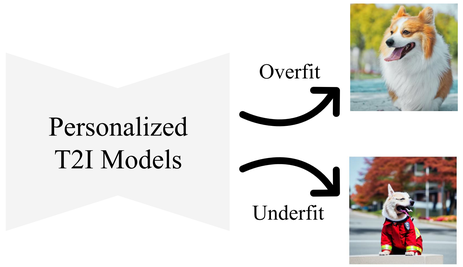
Personalized Text-to-Image Diffusion

- **Integrate novel concepts while preserving prior knowledge**

My dog

- **Challenge: Adaptation–Preservation Trade-off**



- **Preliminary**

1. **Denoising Diffusion Loss (Naïve):**

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{\mathbf{z},\,\mathbf{c},\,\epsilon,\,t}\left\|\epsilon - \epsilon_\theta(\mathbf{z}_t,\,\mathbf{c},\,t)\right\|_2^2$$

2. **Prior Preservation Loss:**

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{denoise}} + \lambda_{\text{prior}}\left\|\epsilon - \epsilon_\theta(\mathbf{z}'_t,\,\mathbf{c}_{\text{class}},\,t)\right\|_2^2\Big]$$

- **Our Contributions**

1. **Objective-level Gap Analysis:** Identification and theoretical analysis of the fundamental mismatch in existing personalization objectives.
2. **Lipschitz-based Regularization:** A simple, theoretically grounded objective guaranteeing distributional preservation during personalization.
3. **Enhanced Performance & Efficiency:** Significant gains in subject fidelity, text alignment, and training-time efficiency, supported by empirical validation.

TL; DR

Is your post-training objective correct?
It requires Lipschitz-based regularization for better stability.

(a) Ideal Case (b) Naïve (c) Prior Preserv. (d) Ours

Key Principle

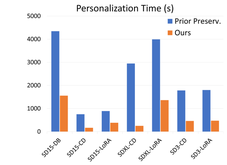
Theorem 2. If the diffusion model ϵ_θ is Lipschitz continuous in θ, for any two parameter sets θ_1 and θ_2, there exists a constant $\lambda > 0$ such that

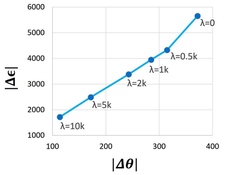
$$D_{KL}(p_{\theta_1}||p_{\theta_2}) \le \lambda \cdot \|\theta_1 - \theta_2\|_k$$

Remark 3. (Simplified) For k=2, $\lambda \cdot \|\theta_1 - \theta_2\|_2 \le \lambda' \cdot \|\theta_1 - \theta_2\|_2^2$

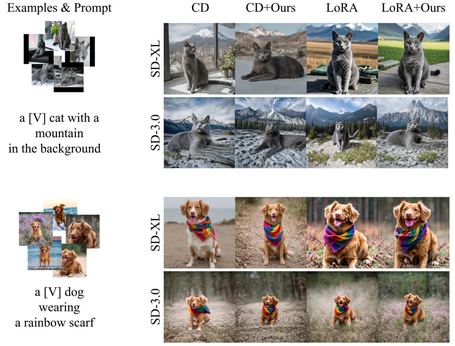
Algorithm 1 Lipschitz-based Regularization for Personalization

Require: Pretrained $\epsilon_{\theta_{\text{base}}}$, dataset $\mathcal{D}_{\text{per}}$, prompt c_{per}, weight λ
Ensure: Personalized $\epsilon_{\theta_{\text{per}}}$
~~**Prior-preservation pre-sampling:**~~
~~for $n = 1, \ldots, N_{\text{cls}}$ do~~
~~$z_{\text{cls}}^{(n)} \sim p_{\epsilon_{\theta_{\text{base}}}}(z \mid c_{\text{cls}})$; $\mathcal{D}_{\text{cls}} \leftarrow \mathcal{D}_{\text{cls}} \cup \{z_{\text{cls}}^{(n)}\}$~~
~~end for~~
$\epsilon_{\theta_{\text{per}}} \leftarrow$ copyWeights($\epsilon_{\theta_{\text{base}}}$); Freeze $\epsilon_{\theta_{\text{base}}}$
Personalization Phase:
for step **do**
  $x \sim \mathcal{D}_{\text{per}}$; $z \leftarrow \mathcal{E}(x)$; $c \leftarrow$ TextEncoder(c_{per})
  $t \sim \text{Unif}(\{1, \ldots, T\})$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\tilde{z}_t := \sqrt{\bar{\alpha}_t}\, z + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$, where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$
  $\mathcal{L} \leftarrow \|\epsilon - \epsilon_{\theta_{\text{per}}}(\tilde{z}_t, t, c)\|_2^2 + \lambda \cdot \sum_i \|\theta_{\text{per}}^i - \theta_{\text{base}}^i\|_2^2$
  Update θ_{per} w.r.t. $\nabla \mathcal{L}$
end for





Experimental Results



Model	Method	DINO ↑	CLIP-T ↑	CLIP-I ↑
SD-1.5	DB	0.6028	0.2793	0.7881
	+ Ours	**0.6394 (+0.0366)**	**0.2976 (+0.0183)**	**0.7948 (+0.0067)**
	CD	0.5568	0.3154	0.7539
	+ Ours	**0.5638 (+0.0070)**	**0.3158 (+0.0004)**	**0.7550 (+0.0011)**
	DB-LoRA	0.5776	**0.3108**	0.7739
	+ Ours	**0.6038 (+0.0262)**	0.2975 (-0.0133)	**0.7915 (+0.0176)**
SD-XL	CD	0.5337	0.3239	0.7485
	+ Ours	**0.5493 (+0.0156)**	**0.3251 (+0.0012)**	**0.7540 (+0.0055)**
	DB-LoRA	0.6562	**0.3099**	0.7970
	+ Ours	**0.6819 (+0.0257)**	0.3014 (-0.0085)	**0.8103 (+0.0133)**
SD-3.0	CD	0.6168	**0.3031**	0.7929
	+ Ours	**0.6271 (+0.0103)**	0.2955 (-0.0076)	**0.8023 (+0.0094)**
	DB-LoRA	0.6043	0.3098	0.7823
	+ Ours	**0.6217 (+0.0174)**	**0.3103 (+0.00005)**	**0.7949 (+0.0126)**

Method	DINO ↑	CLIP-T ↑	CLIP-I ↑	Rank
Few-shot personalization methods				
Ours	**0.6394**	0.2976	**0.7948**	**1**
DreamBooth	0.6028	0.2793	0.7881	3
DreamBooth-LoRA	0.5778	0.3095	0.7731	6
Textual Inversion	0.5342	0.2601	0.7539	9
Custom Diffusion	0.5568	0.3154	0.6479	7
SVDiff	0.3839	**0.3194**	0.6886	8
OFT	0.6320	0.2370	0.7850	4
Conditioning with external resources				
BLIP-Diffusion	0.5943	0.2865	0.7935	5
IP-Adaptor	0.6304	0.2635	0.8318	2